4) Hoff mann's degradation

Amides reacts with bromine in the presence of caustic alkali to form a primary amine carrying one carbon less than the parent amide.

Acetamide — Methyl amine

5) Reduction

Amides on reduction with $LiAlH_4$ or Sodium and ethyl alcohol to form corresponding amines.

Acetamide — Ethyl amine

12.15 Uses of carboxylic acids and its derivatives

Formic acid

It is used

i) for the dehydration of hides.
ii) as a coagulating agent for rubber latex
iii) in medicine for treatment of gout
iv) as an antiseptic in the preservation of fruit juice.

Acetic acid

It is used

i) as table vinegar
ii) for coagulating rubber latex
iii) for manufacture of cellulose acetate and poly vinylacetate

Benzoic acid

It is used

i) as food preservative either in the pure form or in the form of sodium benzoate
ii) in medicine as an urinary antiseptic
iii) for manufacture of dyes

Acetyl Chloride

It is used

i) as acetylating agent in organic synthesis
ii) in detection and estimation of – OH, - NH_2 groups in organic compounds

Acetic anhydride

It is used

i) acetylating agent
ii) in the preparation of medicine like asprin and phenacetin
iii) for the manufacture plastics like cellulose acetate and poly vinyl acetate.

Ethyl acetate is used

i) in the preparation of artificial fruit essences.
ii) as a solvent for lacquers.
iii) in the preparation of organic synthetic reagent like ethyl acetoacetate.

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